

Aakash Kathirvel

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PROFESSIONAL SUMMARY

Data Analytics professional with dual master's degrees in Data Science and Decision Analytics. Experienced in Python, SQL, and visualization tools with a strong foundation in machine learning, predictive modeling, and statistical methods. Skilled at transforming raw data into actionable insights and building scalable analytical solutions that support evidence-based decision making.

EDUCATION

Virginia Commonwealth University, Master of Decision Analytics and Data Science	May 2025
Christ University, India, Master of Science in Data Science	May 2024
Bharathiar University, India, Bachelor of Science in Data Science	May 2023

TECHNICAL SKILLS

- Data Visualization & BI Tools:** Power BI, Tableau, Excel (Pivot Tables, VLOOKUP/HLOOKUP, Power Query, VBA)
- Database Tools :** ETL Pipelines, SQL Joins, Data Cleaning and Transformation
- Statistical & Analytical Methods:** Hypothesis Testing, Regression Modeling, Time Series Forecasting, Optimization, KPI Development, Trend & Variance Analysis
- Cloud & Deployment:** AWS, Azure, Streamlit, Hugging Face Hub.
- Programming Languages:** Python, R, SQL
- Other:** ETL concepts, Data modeling (star schema, DAX), Requirements gathering, Business process mapping

EXPERIENCE

Image Processing Engineer - VCU Convergence Labs Jan 2025 - Present

- Contributed to the development of an interactive Human-Machine Interface (HMI) integrating multi-model AI detection with real-time video streaming and modular control using **Flask, OpenCV, and HTML**.
- Integrated YOLO-World, InsightFace, and DeepFace for **object detection, face recognition, and emotion analysis**, with intelligent voice narration using pyttsx3 and cosine similarity for personalized greetings.
- Designed and optimized a unified AI pipeline enabling on-demand toggling of models via HTML interface, reducing GPU load by **40%** and enhancing response time with smart caching and emotion-state tracking.

Data Analytics Intern - Prodigy Infotech Feb 2024 – Mar 2024

- Conducted end-to-end data analysis using **Google Colab** on real-world datasets including **retail sales, student performance, and traffic accidents**, uncovering actionable insights through visualizations and exploratory techniques.
- Built a **decision tree classifier** to predict customer purchases using demographic and behavioral data, and identified key risk patterns in traffic accidents to support road safety recommendations.

Business Analytics Intern- Hindu Tamil Thisai. Oct 2021 – Dec 2021

- Analyzed customer data related to 15,000+ newspaper subscribers using Excel with VBA and Macros, uncovering reading interest patterns that supported a **5%** increase in subscription renewal strategies.
- Synthesized customer interaction and preference data to provide actionable insights on subscription trends, enabling targeted marketing campaigns through automated Excel dashboards and workflows.

PROJECTS

Global Superstore Sales & Profitability Dashboard using PowerBI June 2025

- Built an interactive **Power BI** dashboard on **51K+** retail transactions using a **star schema** (Fact + DimDate/Customer/Product) and **DAX** (YoY/MoM, rolling-12M, cumulative, contribution); delivered KPIs, monthly trend, geo map, Top-N sub-categories, and discount analysis.
- Implemented advanced UX **drill-through** “Country Details,” **report-page tooltips**, synced slicers, conditional formatting, and filled-map color saturation, surfacing **[X%]** regional margin variance and **[Y]** low-margin sub-categories; reduced ad-hoc reporting time by **[Z]%** for stakeholders.

Cinema Ticket Sales Forecasting using Time Series Models Dec 2024

- Built an end-to-end **time series pipeline in R** (fable/feasts/forecast): cleaned daily cinema ticket sales (date fixes, deduping, interpolation), decomposed with **STL**, established a **TSLM** baseline, and tuned **ARIMA** (manual & auto) and **ETS** models using **AIC** and residual diagnostics to capture weekly seasonality.
- Compared models on a holdout set using **MAPE** and delivered an **ensemble (ARIMA+ETS)** that outperformed single models; produced an **operational 30-day forecast** with clear visualizations to support business decisions.

Office Realignment Optimization

- Developed a binary integer programming model to reassign **80+ stores to 4 regional offices**, reducing annual travel costs by **15%** while balancing employee workloads more efficiently.
- Delivered a feasible solution through Python and AMPL after transitioning from a failed Excel Solver model, leading to a **\$20K** reduction in operational expenses.

Weather Forecasting with LLM

July 2024

- Developed a **Python/Streamlit** web application that integrates the **OpenWeatherMap** API to display real-time KPIs (temperature, humidity, pressure, wind) and a day-wise **5-day/3-hour forecast** with a clean, responsive UI (sidebar city input, metric cards, weekly overview) and robust error handling.
- Implemented **LLM-generated weather summaries** using **OpenAI** to convert raw data into concise insights; organized code into reusable modules and pinned dependencies with **requirements.txt** for reproducible setup, publishing the project on GitHub.

CAMPUS AND COMMUNITY INVOLVEMENT

Coaster Judge volunteer experience with GiSTEM Inc

- Served as Coaster Judge at Kings Dominion Education Days, evaluating K–12 student-built roller coasters on creativity, structural integrity, and performance.
- Promoted STEM engagement by supporting event logistics and fostering a hands-on, educational environment for 100+ students.